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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xuartps.h"
#include "xparameters.h"

XUartPs_Config *uart_config;
XUartPs uart;

void uart_init(){

    uart_config = XUartPs_LookupConfig(XPAR_PS7_UART_1_DEVICE_ID);
    s32 status = XUartPs_CfgInitialize(&uart, uart_config, uart_config->BaseAddress);

    XUartPs_SetBaudRate(&uart, 9600);
    XUartPs_SetOperMode(&uart, XUARTPS_OPER_MODE_NORMAL);

    if(status == XST_SUCCESS)
        xil_printf("UART INIT SUCCESSFUL\n");
    else
        xil_printf("UART INIT FAILED\n");

    while(XUartPs_IsSending(&uart));

}

int main()
{
    u8 data[11] = "Hello World";
    u8 RecvData[11];
    s32 count = 0;
    uart_init();

    XUartPs_SetOperMode(&uart, XUARTPS_OPER_MODE_LOCAL_LOOP);

    XUartPs_Send(&uart, &data[0], 11);
    while(XUartPs_IsSending(&uart));

    while(count != 11) {
        count = count + XUartPs_Recv(&uart, &RecvData[count], 11);
    }

    XUartPs_SetOperMode(&uart, XUARTPS_OPER_MODE_NORMAL);

    int i = 0;

    for(i = 0; i < 11; i++)
    {
        xil_printf("%0c",RecvData[i]);
    }

    cleanup_platform();
    return 0;
}

```